後期長期増強誘導時における可溶性アデニル酸シクラーゼ依存的核cAMP動態

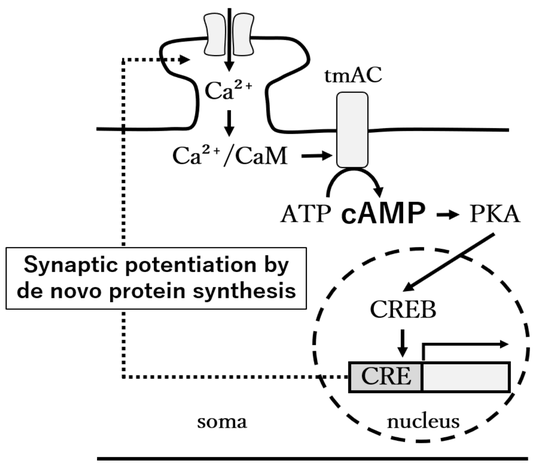
Soluble Adenylyl Cyclase dependent nuclear cAMP dynamics during the induction of Late-phase Long-Term Potentiation.

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Introduction

Long-term potentiation (LTP), a persistent increase in synaptic strength, is thought to serve as an elementary mechanism for learning and memory. Hippocampal LTP is composed of two temporal stages. The Early phase LTP (E-LTP) is a potentiation lasting a few hours independent of de novo protein synthesis. Whereas the Late phase LTP (L-LTP) is a long-lasting potentiation following the E-LTP and requires the protein synthesis activated by 3',5'-cyclic adenosine monophosphate (cAMP) signaling.



The induction of L-LTP is a protein synthesis-dependent process requiring the activation of the cAMP/protein kinase A (PKA) signaling pathway. The canonical model suggests that Ca2+ influx through the N-methyl-D-aspartate (NMDA) receptors activates transmembrane adenylyl cyclase (tmAC) at the plasma membrane. The PKA catalytic subunits migrate into the nucleus and phosphorylate cAMP response element binding protein (CREB). The phosphorylated CREB (pCREB) drives the transcription of immediate early genes (IEGs) for the L-LTP. However, the precise spatiotemporal dynamics of cAMP during L-LTP induction remain poorly understood.

To investigate cAMP dynamics during L-LTP induction, we used the cAMP probe "gCarvi" in cultured mouse hippocampal neurons. We induced L-LTP by applying the high-frequency stimulation (5×HFS; 100 Hz, 1 s, 5 trains) to the hippocampal neurons, which increased pCREB immunoreactivity.

The L-LTP stimulation induced a transient increase in nuclear cAMP (ΔF = 23.2 ± 15.9 %) under physiological bicarbonate buffer conditions, whereas only subtle changes were observed in somatic cAMP. Furthermore, the nuclear cAMP elevation was abolished by the soluble AC (sAC) inhibitor KH7, whereas it was partially inhibited by the tmAC inhibitor SQ22536. Consistent with our previous finding that sAC maintains basal cAMP levels, these results indicate that sAC plays a pivotal role in nuclear cAMP dynamics during L-LTP induction.

Methods

Mouse hippocampal neuron primary cell culture

Hippocampi were dissected from ICR embryos at E17. Hippocampal neurons were dissociated using Neuron Dissociation Solutions (Fujifilm) and plated onto 12 mmΦ coverslips coated with poly-D-lysine (Sigma) and atelocollagen (Koken) at a density of 4×10^4 cells/cm^2 in neurobasal medium supplemented with B27 (Gibco) and 2 mM L-alanyl-L-glutamine (Fujifilm). Dissociated neurons were maintained at 37 °C in 5 % CO_2 for 18-20 DIV. The protocols for animal experiments have been approved by the Doshisha University Animal Research Committee.

Preparation of AAV Vectors

pAAVs, pR2C1, and pHelper (TaKaRa) were co-transfected into AAVpro 293T cells (Clontech) using CalFectin Mammalian Cell Transfection Reagent (SGL). The medium was replaced 6 h after transfection with Dulbecco's modified Eagle's medium (Fujifilm) containing 2 % fetal bovine serum (Gibco). The medium containing virus particles was collected 48 h after the medium replacement and subsequently filtered through a PVDF membrane (Millex-HV, Millipore). The virus solutions were stored at 4 °C until use. rAAV-SynTetOff-NES-rm-gCarvi or rAAV-SynTetOff-rm-gCarvi-3NLS was applied to hippocampal neurons at 4-6 DIV.

Live cell imaging

Experiments were performed in a Custom balanced salt solution (120 mM NaCl, 26.2 mM $NaHCO_3$, 3 mM KCl, 2 mM $CaCl_2$, 0.5 mM $MgCl_2$, 0.5 mM $MgSO_4$, 1 mM Na_2HPO_4 , 1 mM NaH_2PO_4, 5.5 mM D-Glucose, 1 mM Lactate Na and 10 mM HEPES) with 10 μM muscimol unless otherwise noted. All images were taken using a confocal microscopy (LSM710, Zeiss) with a 20×, 0.8 N.A. dry objective lens. Argon 488 nm and DPSS 561 nm were used for excitation of cpGFP and mCherry, respectively. Emissions were collected between 496 nm and 535 nm for cpGFP, and 600 nm and 700 nm for mCherry. Time-lapse images were acquired every 6 s using Zen 2009 software (Zeiss).

Parallel platinum wires (Unique Medical) were placed on the coverslips then connected to an electronic stimulator (15 V, 200 μs duration, 5 s interval, SEN-7203, Nihon Kohden) for field electrical stimulation (100 Hz, 1 s, 1–5 trains).

Conclusion

5×HFS induces a transient increase in somatic and nuclear cAMP, and somatic PKA activity in cultured hippocampal neurons (Fig. 1, 2). These cAMP increases depend on action potential and on activation of both GluN and GluA, indicating that it is triggered via excitatory synaptic input (Fig. 3). Both sAC and tmAC contribute to these cAMP increases: sAC inhibition abolished the responses, whereas tmAC inhibition only partially suppressed them (Fig. 4). In contrast, inhibition of sAC or tmAC had little effect on 5×HFS-induced somatic PKA activation (Fig. 5).

These results suggest that 5×HFS-induced somatic PKA activation may be driven by postsynaptically compartmentalized cAMP signaling.

Disclosure of Conflict of Interest
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Results

5×HFS Induces a Transient Increase in Somatic and Nuclear cAMP

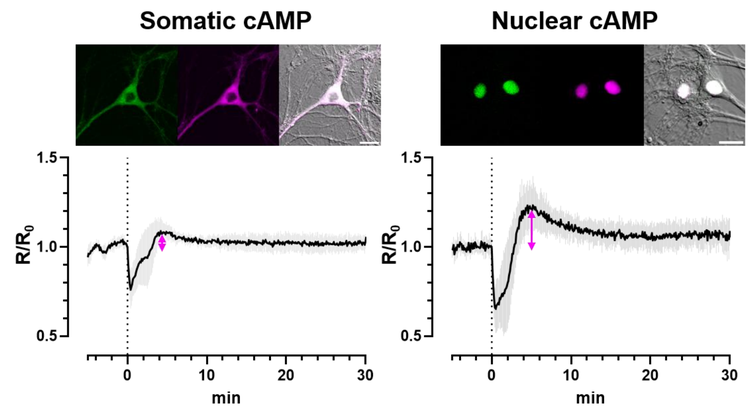


Fig.1. Transient increase in somatic and nuclear cAMP following L-LTP-inducing stimulation (5×HFS). Cultured hippocampal neurons expressing NES-gCarvi (left, n = 12) or gCarvi-3NLS (right, n = 13) were subjected to field electrical stimulation, and changes in fluorescence ratio were monitored (mean ± S.D.).

5×HFS Induces Transient PKA Activation in the Soma

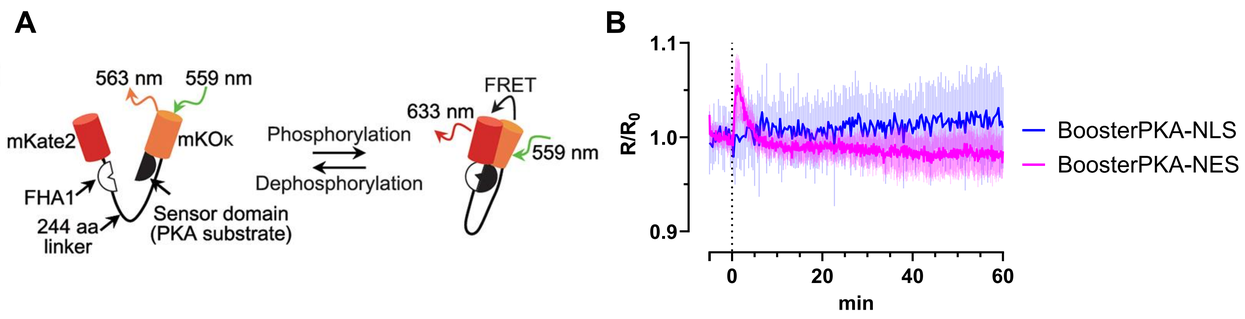


Fig. 2. PKA activity in the soma (magenta) and nucleus (blue) was measured using the FRET-based PKA activity sensor BoosterPKA following 5×HFS (mean ± S.D.). (A) Cartoon of BoosterPKA. (Adapted from Watabe T. et al., 2020, *ACS Sens.*) (B) Measurements taken every 20 s for 60 min post-stimulation.

TTX, CNQX, and D-APV Block the 5×HFS-Induced cAMP Increase

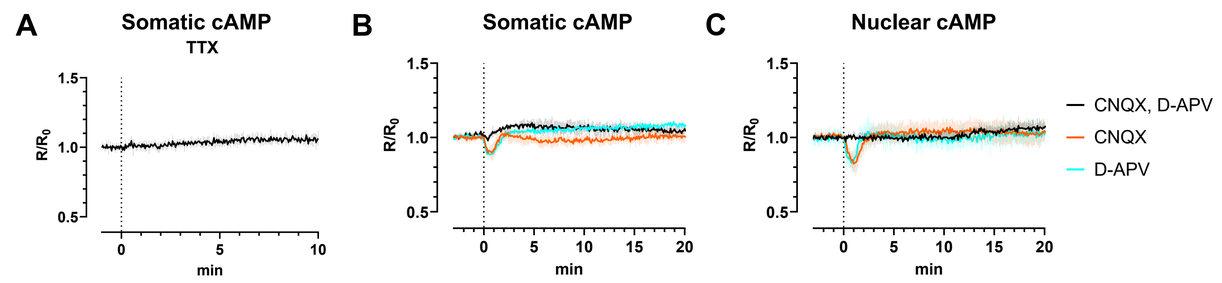


Fig. 3. cAMP dynamics during 5×HFS in the presence of inhibitors of voltage-gated Na^+ channels and NMDA/non-NMDA glutamate receptors (mean ± S.D.). (A) The somatic cAMP increase evoked by 5×HFS was completely abolished in the presence of 1 μM tetrodotoxin (TTX) (n = 6). (B) Glutamate receptor antagonists blocked the somatic cAMP response to 5×HFS (10 μM CNQX + 50 μM D-APV, n = 3; 10 μM CNQX, n = 5; 50 μM D-APV, n = 3). (C) Glutamate receptor antagonists similarly blocked the nuclear cAMP response to 5×HFS (10 μM CNQX + 50 μM D-APV, n = 6; 10 μM CNQX, n = 8; 50 μM D-APV, n = 9).

Inhibition of sAC or tmAC Eliminates the 5×HFS-Induced cAMP Increase

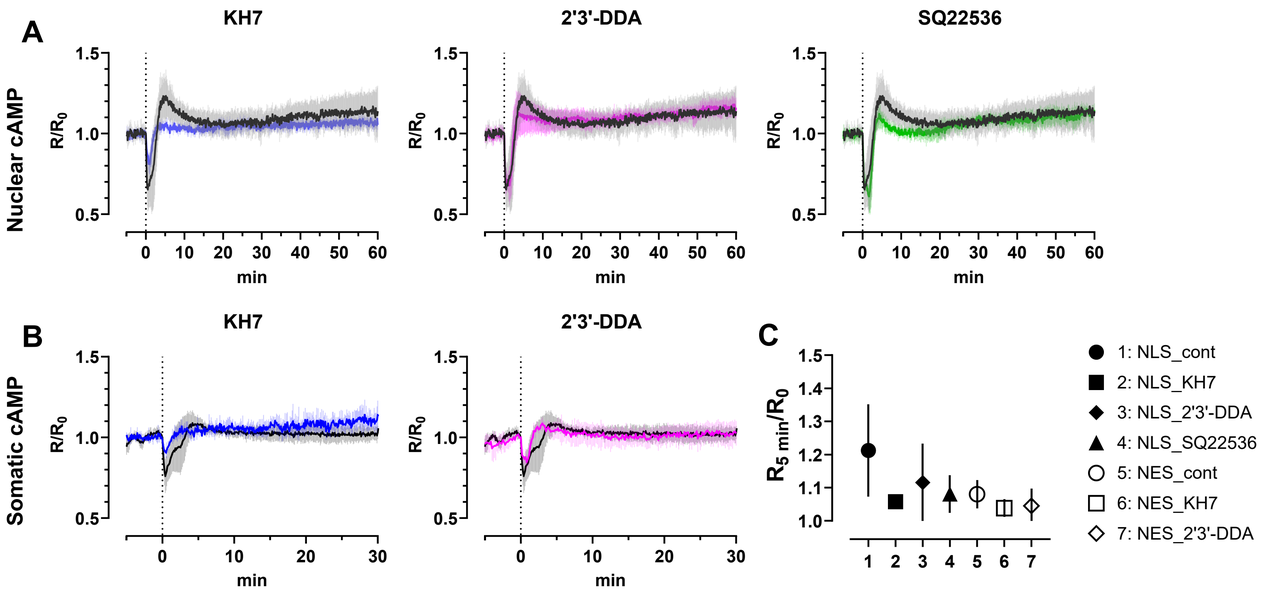


Fig. 4. Nuclear (A) and somatic (B) cAMP dynamics during 5×HFS in the presence of adenylyl cyclase inhibitors (mean ± S.D.). The sAC-specific inhibitor KH7 (10 μM) and the tmAC-specific inhibitors 2′,3′-dideoxyadenosine (2′,3′-DDA) and SQ22536 (100 μM each) were used. (C) Fluorescence ratio changes at 5 min post-stimulation. NLS_cont = 1.21 ± 0.13 (n = 13), NLS_KH7 = 1.06±0.02 (n = 4), NLS_2'3'-DDA = 1.12 ± 0.11 (n = 13), NLS_SQ22536 = 1.08 ± 0.05 (n = 6), NES_cont = 1.08 ± 0.04 (n = 12), NES_KH7 = 1.04 ± 0.02 (n = 3), NES_2'3'-DDA = 1.05 ± 0.05 (n = 5).

Inhibition of sAC or tmAC Does Not Affect Somatic PKA Activation

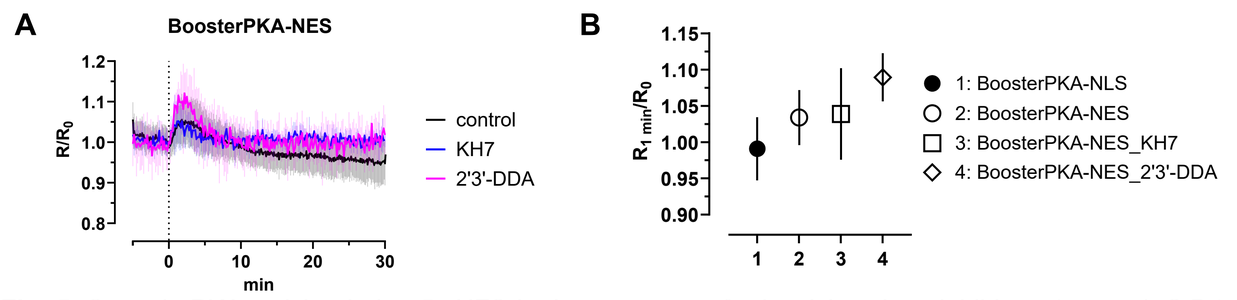


Fig. 5. Somatic PKA activity during 5×HFS in the presence of adenylyl cyclase inhibitors (mean ± S.D.). (A) Time course of PKA activity during 5×HFS in the absence (control) or presence of the sAC-specific inhibitor KH7 (10 μM) or the tmAC-specific inhibitor 2′,3′-DDA (100 μM), measured every 6 s as in Fig. 2A. (B) FRET ratio changes at 1 min post-stimulation. BoosterPKA-NLS = 0.99 ± 0.04 (n = 11), BoosterPKA-NES = 1.03 ± 0.04 (n = 29), BoosterPKA-NES_KH7 = 1.04 ± 0.05 (n = 3), BoosterPKA-NES_2′,3′-DDA = 1.09 ± 0.04 (n = 7).